Smart Metro Ticket Booking System

Project Development Phase

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

1. User Acceptance Testing

1. Purpose

This document sets out the User Acceptance Testing (UAT) approach for the metro digital ticketing solution built on ServiceNow. UAT confirms the solution meets business requirements from the standpoint of Passengers, Station Managers, the Metro Operations Team, and IT Admins before production go-live is approved.

2. UAT Scope & Objectives

- Validate end-to-end ticket booking: Source, Destination, Passenger Type, and Number of Tickets selection through the Service Catalog.
- Confirm automated fare calculation, including concessions for Student and Senior Citizen passenger types.
- Verify digital payment processing via UPI, credit/debit cards, and mobile wallets.
- Confirm QR code generation, delivery, and validation at AFC gates for entry and exit.
- Validate Station Details Mapping, reporting dashboards, and role-based access for Station Managers, Operations Team, and IT Admins.
- Confirm correct behaviour across all channels: Mobile App, ServiceNow Portal, and WhatsApp.

3. UAT Process Flow

UAT follows a structured six-step process, from planning through stakeholder sign-off, with a retest loop for any defects found during execution.

Figure 1 — UAT Process Flow

4. UAT Test Scenarios

Scenario ID	Scenario Description	Related Milestone
UAT-SC-01	Passenger books a ticket for a single trip using General passenger type.	Catalog Form Setup
UAT-SC-02	Passenger books multiple tickets in a single transaction.	Catalog Form Setup
UAT-SC-03	System applies Student/Senior Citizen concession correctly.	Fare Logic Script
UAT-SC-04	Passenger completes payment via UPI, card, and wallet.	Fare Logic Script
UAT-SC-05	QR code is generated and delivered after successful payment.	QR Code Integration
UAT-SC-06	QR code is validated correctly at AFC gate for entry and exit.	QR Code Integration
UAT-SC-07	Station Manager views real-time bookings for assigned station.	Station Details Mapping
UAT-SC-08	Passenger completes booking via WhatsApp channel.	UI Pages
UAT-SC-09	Operations Team updates fare/route configuration centrally.	Station Details Mapping

5. Sample UAT Test Cases

Test Case ID	Test Steps	Expected Result
TC-01	Open catalog form; select Source, Destination, Passenger Type = General, Number of Tickets = 1; submit.	Fare is displayed correctly and booking proceeds to payment.
TC-02	Repeat booking with Passenger Type = Student.	Discounted fare is applied automatically per the concession policy.
TC-03	Complete payment using UPI on a successful test transaction.	Payment is confirmed and QR code is generated within 5 seconds.
TC-04	Scan generated QR at AFC entry gate simulator.	Gate validates QR as valid and logs entry; QR status updates to 'Scanned at Entry'.
TC-05	Attempt to reuse the same QR code for a second entry.	System rejects the scan as invalid/reused and logs the attempt.
TC-06	Enter Source and Destination as the same station.	System blocks submission with a validation message.

6. Defect Severity Classification

Defects found during UAT execution are classified by severity to prioritize fixes. The chart below shows an illustrative distribution used for planning retest effort; actual figures will be tracked once execution begins.

Figure 2 — Illustrative UAT Defect Severity Distribution

7. UAT Team & Roles

Role	Responsibility
Passengers (Test Users)	Execute booking, payment, and gate scenarios from an end-user perspective.
Station Managers	Validate station-level dashboards, gate activity, and exception handling.
Metro Operations Team	Validate fare/route configuration and reporting accuracy.
IT Admins	Support environment setup, defect triage, and integration verification.
QA / Test Lead	Own the UAT plan, coordinate execution, and track defects to closure.

8. Entry & Exit Criteria

8.1 Entry Criteria

- System Integration Testing (SIT) is complete with no open Critical/High defects.
- UAT environment is configured with representative station and fare data.
- Test scenarios and test cases are reviewed and approved by stakeholders.
- UAT participants from all stakeholder groups are identified and available.

8.2 Exit Criteria

- All planned UAT test cases have been executed.
- No open Critical or High severity defects remain.
- Any open Medium/Low defects have documented workarounds and stakeholder agreement.
- Formal sign-off obtained from Passengers' representative, Station Managers, Operations Team, and IT Admins.

9. Sign-off

Once UAT is completed successfully against the exit criteria above, stakeholders provide formal written sign-off, after which the solution moves to the Deployment milestone as defined in the Project Planning Phase document.